

Symmetry Breaking and Phase Transition in $Z_2 \times Z_2$ Invariant Two Scalar Model

Junichi Haruna
Kyoto Univ.

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Based on JH, H Kawai arXiv:1905.05656 [hep-th],
and works in progress with H Kawai & K Sakai

Introduction

- ◆ Phase structure of phi fourth theory is well-known.

$$V = m^2\phi^2 + \lambda\phi^4$$

... broken/unbroken phases depend on the sign of m^2 .

- ◆ Phase structure of two scalar theory is **unknown**.

$$V = m_1^2\phi^2 + m_2^2\psi^2 + \rho\phi^4 + \rho'\psi^4 + \kappa\phi^2\psi^2$$

- ◆ In 4 dim, we found that in the **massless** case **one scalar field** have a vacuum expectation value:

$$V = \rho\phi^4 + \rho'\psi^4 + \kappa\phi^2\psi^2 \Rightarrow \langle\phi\rangle \neq 0, \langle\psi\rangle = 0$$

- ◆ This implies two scalar theory has **non-trivial phase structure**.

Introduction

- ◆ We study $O(N) \times O(N)$ scalar model in 3 or 4 dimensions:

$$\mathcal{L} = N \left[\frac{1}{2} (\partial_\mu \phi_i)^2 + \frac{1}{2} (\partial_\mu \psi_i)^2 + \frac{m_0^2}{2} \phi_i^2 + \frac{m_0'^2}{2} \psi_i^2 \right. \\ \left. + \frac{\rho_0}{8} (\phi_i^2)^2 + \frac{\kappa_0}{4} \phi_i^2 \psi_i^2 + \frac{\rho_0'}{8} (\psi_i^2)^2 \right]$$

$(i = 1, \dots, \underline{\underline{N}})$

- ◆ For $N = 1$, This model corresponds to $Z_2 \times Z_2$ invariant two scalar model.
- ◆ In the large N limit ($N \rightarrow \infty$), we can calculate the effective potential **exactly, including all-order loop effects.**

Summary

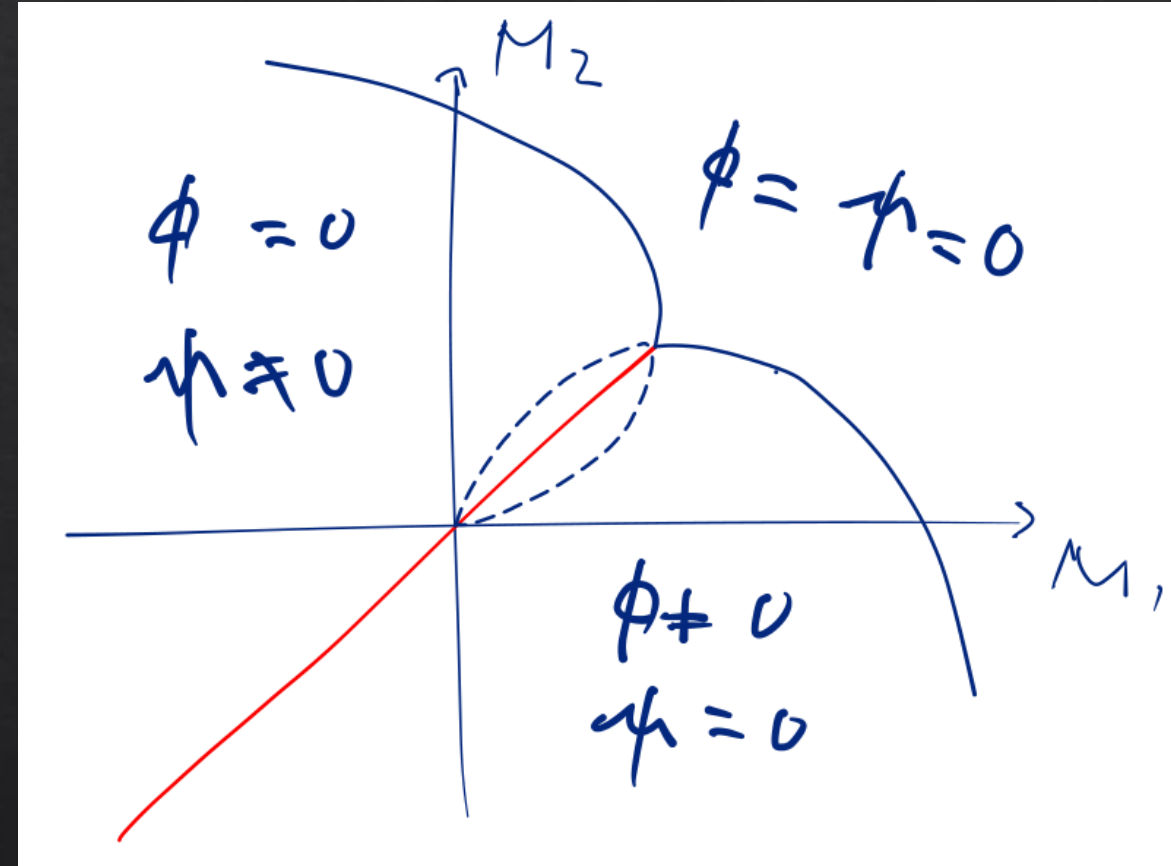
◇ Today, I talk about ...

1. Phase diagram in 3 dim.

- $0 < \rho_0 < \kappa_0$

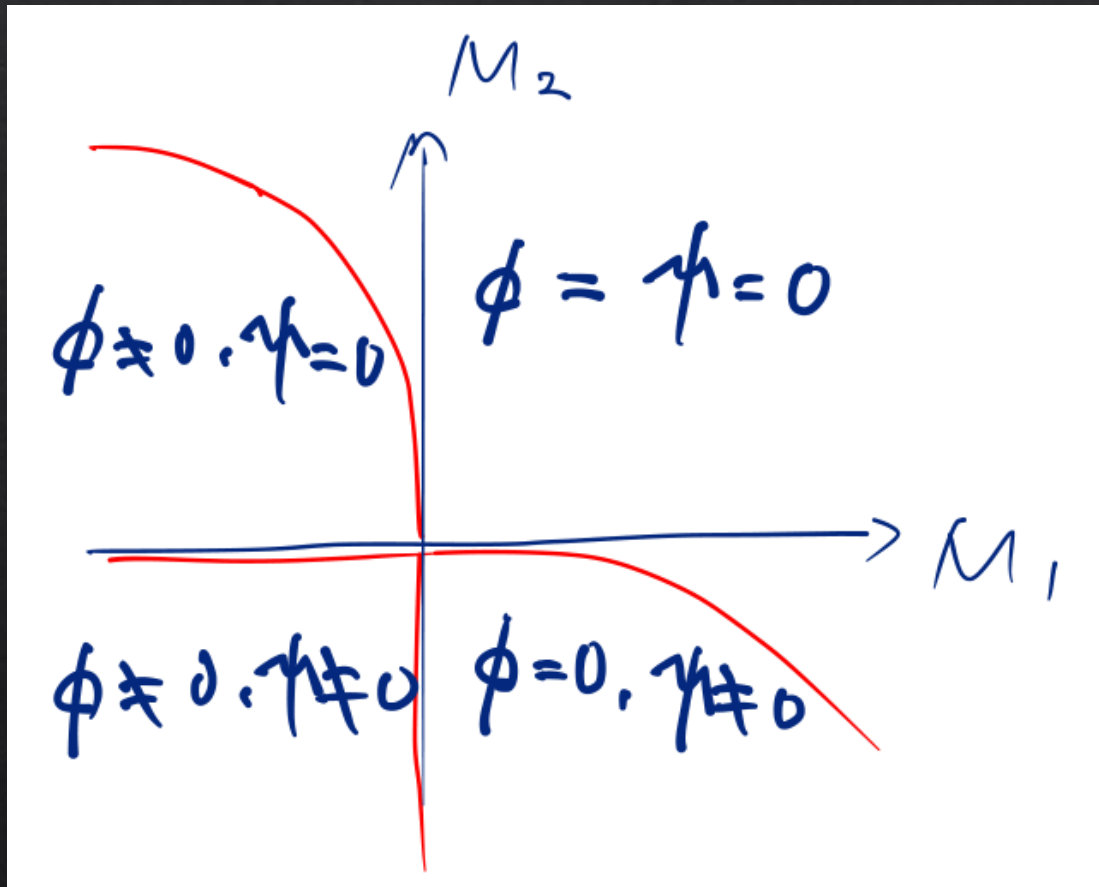
$$(M_i := (\lambda^{-1} m^2)_i, \lambda^{-1} := \begin{pmatrix} \rho_0 & \kappa_0 \\ \kappa_0 & \rho_0 \end{pmatrix})$$

(We take $\rho_0 = \rho'_0$ for simplicity)

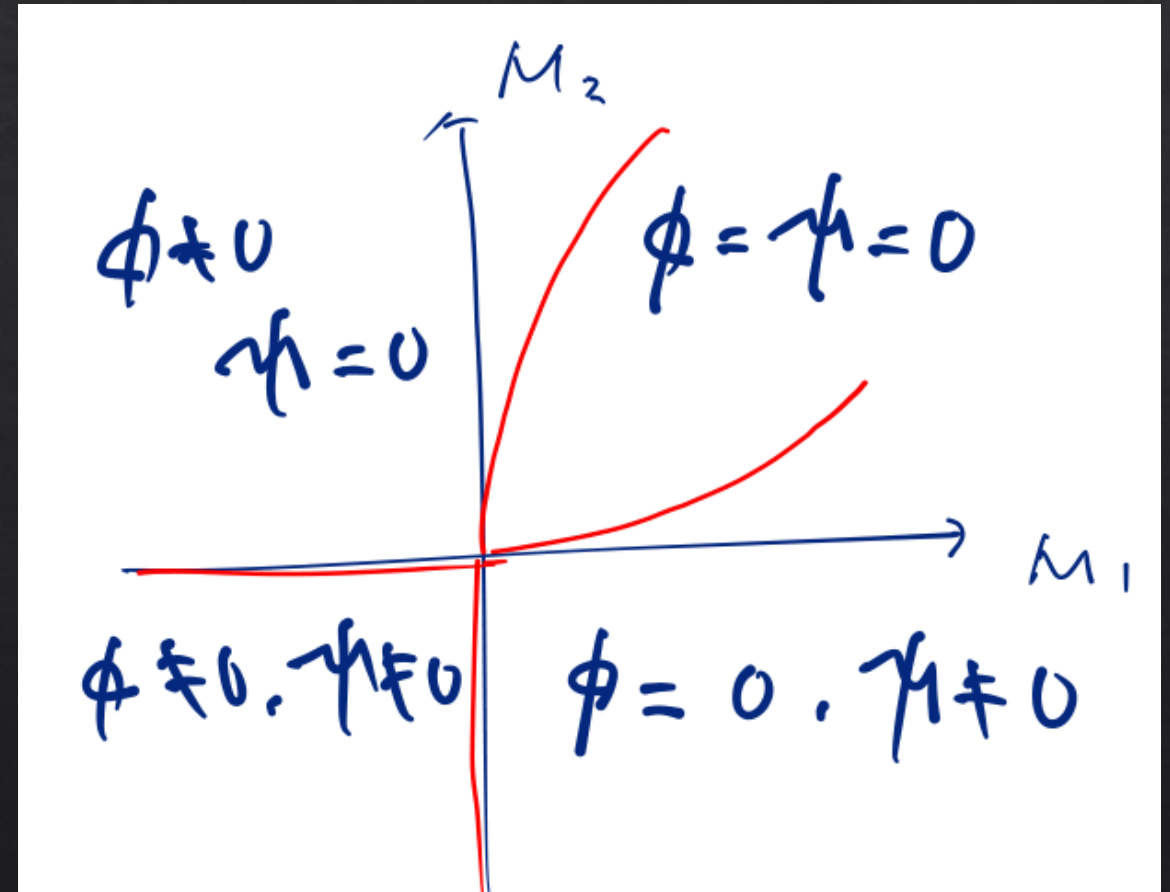


Summary

- $0 < \kappa_0 < \rho_0$



- $\kappa_0 < 0 < \rho_0$



Summary

2. The massless ($m_1 = m_2 = 0$) case in 4 dim:

$$\langle \psi \rangle^2 = 0, \langle \phi \rangle^2 = \mu_*^2 \exp \left(-32\pi^2 \lambda_{22}^{-1}(\mu_*) \right)$$

$$(\mu_* \cdots \rho(\mu_*) = 0)$$

◊ This can be applied for dynamical generation of Weak scale.

(SM + two SM singlet scalars (Dark Matter?))

$$\begin{aligned} \mathcal{L} = & \left| \partial_\mu H \right|^2 - \frac{\eta}{2} \phi^2 |H|^2 + \frac{\eta'}{2} \psi^2 |H|^2 + \lambda |H|^4 \\ & + \frac{1}{2} (\partial_\mu \phi)^2 + \frac{1}{2} (\partial_\mu \psi)^2 + \frac{\rho}{4!} \phi^4 + \frac{\kappa}{4} \phi^2 \psi^2 + \frac{\rho'}{4!} \psi^4 \end{aligned}$$

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Model

- ◊ We study $O(N) \times O(N)$ invariant two scalar model:

$$\mathcal{L} = N \left[\frac{1}{2} (\partial_\mu \phi_i)^2 + \frac{1}{2} (\partial_\mu \psi_i)^2 + \frac{m_0^2}{2} \phi_i^2 + \frac{m_0'^2}{2} \psi_i^2 \right. \\ \left. + \frac{\rho_0}{8} (\phi_i^2)^2 + \frac{\kappa_0}{4} \phi_i^2 \psi_i^2 + \frac{\rho_0'}{8} (\psi_i^2)^2 \right]$$

$(i = 1, \dots, \underline{\underline{N}})$

- ◊ We calculate the effective potential in $N \rightarrow \infty$ limit.
- ◊ Point: rewriting the tree potential in **quadratic form of fields, introducing auxiliary fields.**

Calculation of Effective potential

◊ Rewriting the tree-level potential:

$$\frac{\mathcal{L}}{N} = (kin.) + \underbrace{\frac{1}{2} (m_0^2 \ m_0'^2)}_{M_0^{2t}} \underbrace{\begin{pmatrix} \phi_i^2 \\ \psi_i^2 \end{pmatrix}}_{\Phi^2} + \frac{1}{8} \begin{pmatrix} \phi_i^2 \\ \psi_i^2 \end{pmatrix}^t \underbrace{\begin{pmatrix} \rho_0 & \kappa_0 \\ \kappa_0 & \rho_0' \end{pmatrix}}_{\lambda} \begin{pmatrix} \phi_i^2 \\ \psi_i^2 \end{pmatrix}$$

$$= (kin.) + \frac{1}{2} M_0^{2t} \Phi^2 + \left(\frac{\lambda}{2} \Phi^2 \right)^t \frac{\lambda^{-1}}{2} \left(\frac{\lambda}{2} \Phi^2 \right)$$

$$= (kin.) + \left(\frac{\lambda}{2} \Phi^2 + M_0^2 \right)^t \frac{\lambda^{-1}}{2} \left(\frac{\lambda}{2} \Phi^2 + M_0^2 \right)$$

$$= (kin.) - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} \left(\frac{\lambda}{2} \Phi^2 + M_0^2 \right)$$

$C := \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$: auxiliary fields

Calculation of Effective potential

$$\frac{\mathcal{L}}{N} = \frac{1}{2} \phi_i (-\Delta^2 + c_1) \phi_i + \frac{1}{2} \psi_i (-\Delta^2 + c_2) \psi_i - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M^2$$

◊ We set $\phi_i(x) = \phi \delta_{i,1} + \widehat{\phi}_i(x)$, $\psi_i(x) = \psi \delta_{i,1} + \widehat{\psi}_i(x)$

and drop the linear terms of $\widehat{\phi}_i(x)$, $\widehat{\psi}_i(x)$

$$= \frac{1}{2} \widehat{\phi}_i (-\Delta^2 + c_1) \widehat{\phi}_i + \frac{1}{2} \widehat{\psi}_i (-\Delta^2 + c_2) \widehat{\psi}_i - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

$$\therefore \int Dc D\phi_i D\psi_i \exp(-S)$$

$$\propto \int Dc \exp \left(-N \int d^4x \sum_i \frac{\text{tr}}{2} \log(-\Delta^2 + c_i) - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix} \right)$$

Calculation of Effective potential

$$\int Dc D\phi_i D\psi_i \exp(-S) \propto \int Dc \exp\left(-N \int d^4x \sum_i \frac{\text{tr}}{2} \log(-\Delta^2 + c_i) - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M_0^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}\right)$$

◊ C integral is equivalent to **substituting the value of the stationary point** for C in the large N limit:

$$\exp\left(-\int d^4x V_{\text{eff}}\right) \propto \int Dc D\phi_i D\psi_i \exp(-S) \propto \exp\left(-N \int d^4x \sum_i \frac{\text{tr}}{2} \log(-\Delta^2 + c_i) - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M_0^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}\right)$$

Calculation of Effective potential

$$\therefore \frac{V_{\text{eff}}}{N} = \sum_i \frac{\text{tr}}{2} \log(-\Delta^2 + c_i) - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M_0^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

◊ The values of c_i are determined by $\partial_{c_i} V_{\text{eff}} = 0$

◊ Using $\frac{\text{tr}}{2} \log(-\Delta^2 + c_i) = \frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} \log(k^2 + c_i) = \frac{\Lambda}{4\pi} c_i - \frac{\sqrt{c_i}^3}{12\pi} + O\left(\frac{1}{\Lambda}\right)$

$$\frac{V_{\text{eff}}}{N} = \frac{\Lambda}{4\pi} c_i - \frac{\sqrt{c_i}^3}{12\pi} - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M_0^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

◊ We need to renormalize the linear divergence of Λ :

$$\frac{\Lambda}{4\pi} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda^{-1} M_0^2 =: \lambda^{-1} M^2$$

Vacuum

$$\therefore \frac{V_{\text{eff}}}{N} = -C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix} - \sum_i \frac{\sqrt{c_i}^3}{12\pi}$$

◆ The extreme condition is

$$0 = \begin{pmatrix} \partial_\phi \\ \partial_\psi \end{pmatrix} V_{\text{eff}} = \begin{pmatrix} \widetilde{\partial_\phi} \\ \widetilde{\partial_\psi} \end{pmatrix} V_{\text{eff}} = \begin{pmatrix} c_1 \phi \\ c_2 \psi \end{pmatrix}$$

the derivative of explicit ϕ, ψ dependence
remains from $\partial_{c_i} V_{\text{eff}} = 0$.

◆ In order to find minimum, we solve $\partial_{c_i} V_{\text{eff}} = 0$ in the cases

$$(1) c_1 = c_2 = 0 \quad (2) c_1 = \psi = 0, c_2 \neq 0$$

$$(2') c_2 = \phi = 0, c_1 \neq 0 \quad (3) \phi = \psi = 0, c_1 \neq 0, c_2 \neq 0$$

and compare the value of V_{eff} .

Phase structure in 3 dim

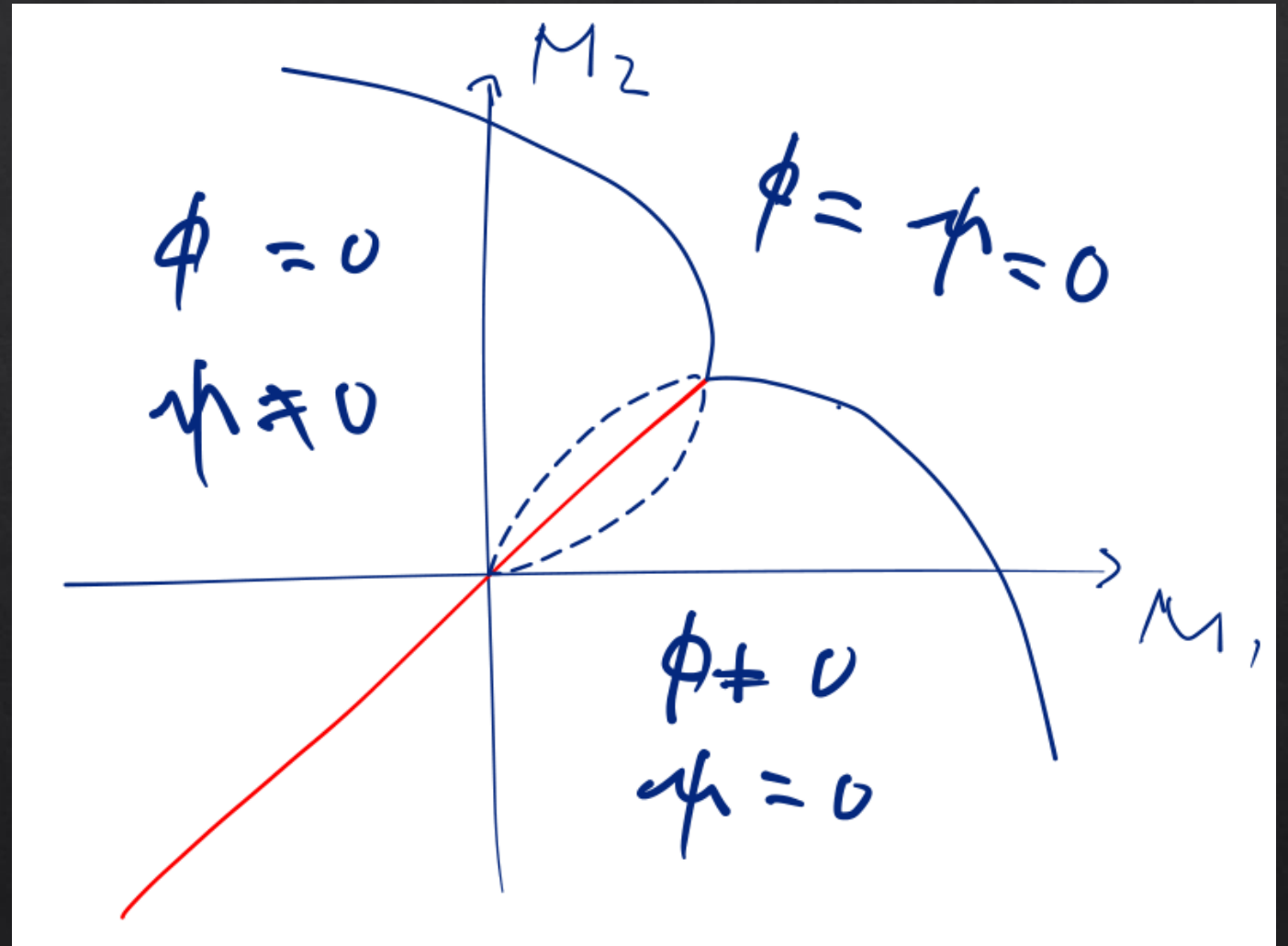
◆ Results

- $0 < \rho_0 < \kappa_0$

$$(M_i := (\lambda^{-1} m^2)_i)$$

$$\lambda^{-1} := \begin{pmatrix} \rho_0 & \kappa_0 \\ \kappa_0 & \rho_0 \end{pmatrix})$$

(we take $\rho_0 = \rho'_0$
for simplicity)



Phase structure in 3 dim

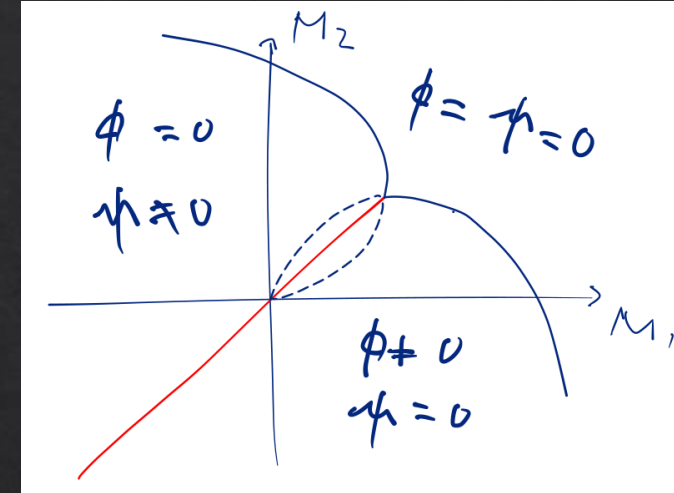
◇ Comments ($0 < \rho_0 < \kappa_0$):

1. There is **no phase** where vacuum expectation values of **both fields are nonzero**.

← $\kappa\phi^2\psi^2$ term in the potential prevents it.

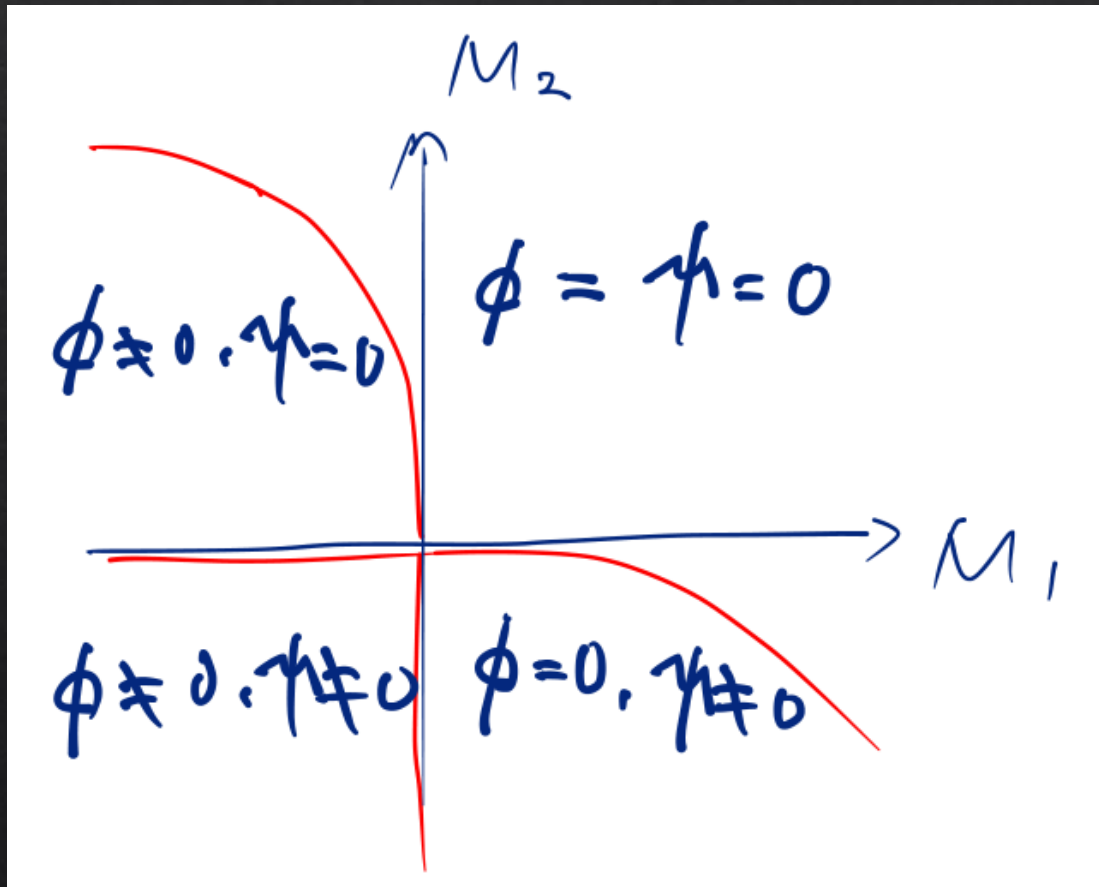
2. **One broken phase** exist in a region where **mass squares are positive**.

← because of the loop contribution to the effective potential?

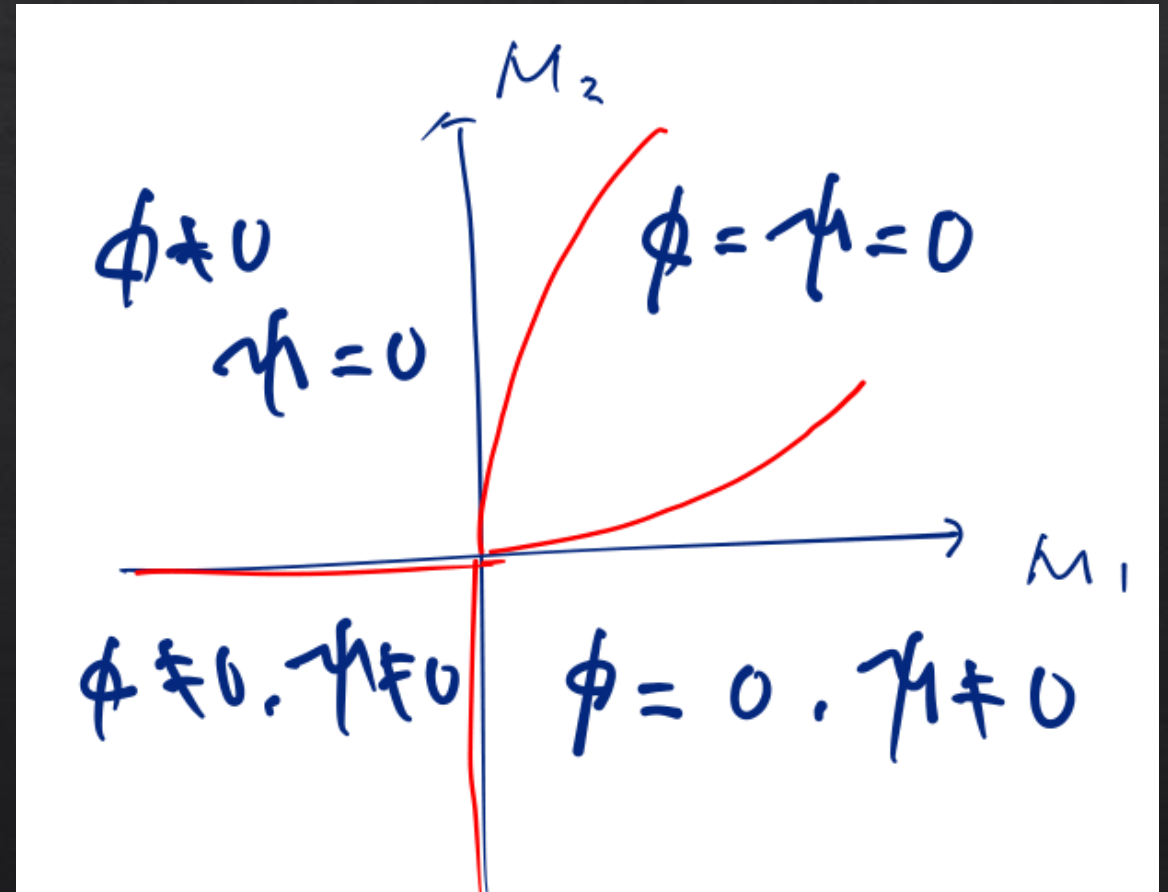


Phase structure in 3 dim

- $0 < \kappa_0 < \rho_0$



- $\kappa_0 < 0 < \rho_0$



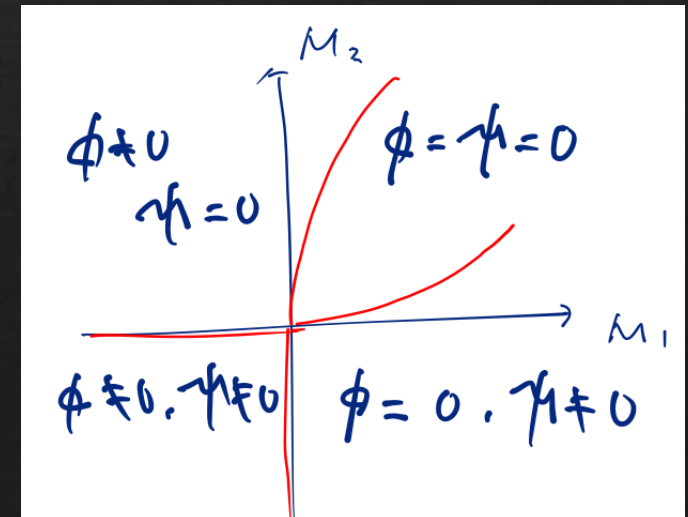
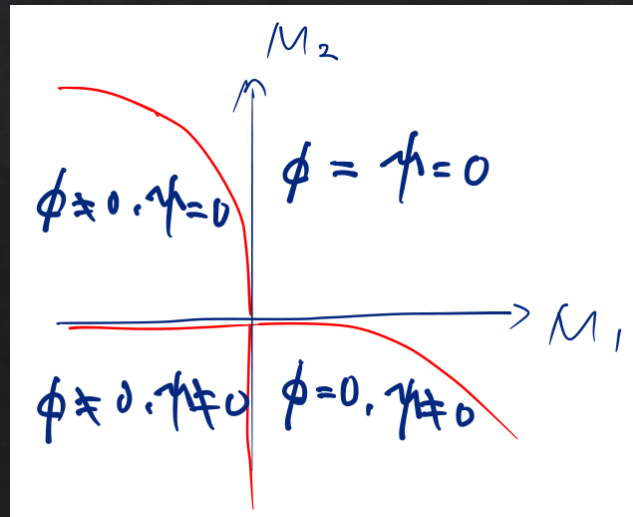
Phase diagram in 3 dim

◆ Comments ($\kappa_0 < \rho_0$):

1. The phase diagram are essentially **same** as **phi fourth theory**.
2. The boundaries between **both broken** and **one broken** phases ($M_1 = 0, M_2 < 0$ and $(1 \leftrightarrow 2)$) are the boundary where the **potential can be negative**:

$$V = \frac{\lambda^{-1}}{2} \left(\frac{\lambda}{2} \Phi^2 + m^2 \right)^2$$

$$= \frac{\lambda}{2} \left(\frac{1}{2} \Phi^2 + M \right)^2$$



Massless $O(N) \times O(N)$ invariant model in 4 dim

◊ In 4 dim, we have not analyzed the entire phase structure.

◊ We studied **massless** $O(N) \times O(N)$ invariant model in **4 dim**.

(the renormalized masses are zero: $m_1^2 = m_2^2 = 0$)

$$\mathcal{L} = N \left[\frac{1}{2} (\partial_\mu \phi_i)^2 + \frac{1}{2} (\partial_\mu \psi_i)^2 + \frac{\rho}{8} (\phi_i^2)^2 + \frac{\kappa}{4} \phi_i^2 \psi_i^2 + \frac{\rho'}{8} (\psi_i^2)^2 + (\text{counter terms}) \right]$$

◊ As in 3 dim, the effective potential can be calculated exactly in large N limit.

◊ (Here, we take $\rho < \rho'$ without loss of generality.)

Effective potential in 4 dim

◊ Calculation is the same as 3 dim,
but the momentum integral differs:

$$\text{tr} \log(-\Delta^2 + c_i) = \int \frac{d^4 k}{(2\pi)^4} \log(k^2 + c_i) = \frac{c_i^2}{32\pi^2} \left(\log \frac{c_i}{\Lambda^2} - \frac{1}{2} \right) + \frac{c_i}{32\pi^2} \Lambda^2$$

◊ The effective potential is

$$\frac{V_{\text{eff}}}{N} = \sum_i \frac{c_i^2}{64\pi^2} \left(\log \frac{c_i}{\Lambda^2} - \frac{1}{2} \right) + \frac{c_i}{32\pi^2} \Lambda^2 - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M_0^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

Effective potential in 4 dim

$$\frac{V_{\text{eff}}}{N} = \sum_i \frac{c_i^2}{64\pi^2} \left(\log \frac{c_i}{\Lambda^2} - \frac{1}{2} \right) + \frac{c_i}{32\pi^2} \Lambda^2 - C^t \frac{\lambda^{-1}}{2} C + C^t \lambda^{-1} M_0^2 + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

◆ We need to renormalize the quadratic and log divergence of Λ

$$\begin{cases} \frac{\Lambda^2}{32\pi^2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda^{-1} M_0^2 = 0 \\ \frac{1}{64\pi^2} \log \frac{\mu^2}{\Lambda^2} I_2 - \frac{\lambda^{-1}}{2} = -\frac{\lambda^{-1}(\mu)}{2} \end{cases}$$

Here, we consider the case where renormalized masses are 0

∴

$$\frac{V_{\text{eff}}}{N} = \sum_i \frac{c_i^2}{64\pi^2} \left(\log \left(\frac{c_i}{\mu^2} \right) - \frac{1}{2} \right) - C^t \frac{\lambda^{-1}}{2} C + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

Vacuum

$$\frac{V_{\text{eff}}}{N} = \sum_i \frac{c_i^2}{64\pi^2} \left(\log\left(\frac{c_i}{\mu^2}\right) - \frac{1}{2} \right) - C^t \frac{\lambda^{-1}}{2} C + \frac{1}{2} C^t \begin{pmatrix} \phi^2 \\ \psi^2 \end{pmatrix}$$

- ◆ After comparing the values of V_{eff} , we find the minimum point is

$$\psi^2 = 0, \phi^2 = \mu^2 \exp\left(-32\pi^2 \lambda_{22}^{-1}\right)$$

$$(c_1 = 0, c_2 = 2\lambda_{12}^{-1} \mu^2 \exp\left(-32\pi^2 \lambda_{22}^{-1}\right))$$

- ◆ Can we use this to explain the origin of Weak scale,
by coupling the model to the standard model?

Weak Scale

- ◆ Let us change the Higgs Sector of SM:

$$\mathcal{L} = |\partial_\mu H|^2 - \frac{\eta}{2} \phi^2 |H|^2 + \frac{\eta'}{2} \psi^2 |H|^2 + \lambda |H|^4 \\ + \frac{1}{2} (\partial_\mu \phi)^2 + \frac{1}{2} (\partial_\mu \psi)^2 + \frac{\rho}{4!} \phi^4 + \frac{\kappa}{4} \phi^2 \psi^2 + \frac{\rho'}{4!} \psi^4$$

- ◆ ϕ and ψ are **SM singlet** scalars.
- ◆ **The vacuum expectation value of ϕ gives the negative mass term of Higgs fields:**

$$-\frac{\eta}{2} \langle \phi \rangle^2 |H|^2 + \lambda |H|^4 \simeq -\frac{m_H^2}{2} |H|^2 + \lambda |H|^4$$

= Weak scale ($\langle H \rangle = 246 \text{ GeV}$) is reproduced dynamically!

Conclusion

- ◈ We studied $O(N) \times O(N)$ scalar model in 3 or 4 dimensions in the large N limit.
- ◈ In 3 dim, it has non-trivial phase structure.
- ◈ In 4 dim, we have not studied it completely yet,
but one scalar field has a vacuum expectation value even in the massless case.
- ◈ This can be used for dynamical generation of Weak scale.

Future Works

- ◊ We continue to study the phase structure in 4 dim.
and order of phase transition.
- ◊ Models with more than 2 scalar fields are interesting.
- ◊ If the $\frac{1}{N}$ correction is considered, what does it become?

These results are universal or peculiar to the large N limit?

Thank you!

Weak Scale

◇ ϕ, ψ の質量

$$m_{\phi}^2 = \left. \frac{d^2}{d\phi^2} V_{\text{eff}} \right|_{\phi=\langle\phi\rangle} = \frac{\kappa^2(\mu_*)}{32\pi^2} \langle\phi\rangle^2$$

$$m_{\psi}^2 = \frac{\kappa(\mu_*)}{2} \langle\phi\rangle^2$$

◇ ϕ, ψ は何者?

Higgsの真空期待値と質量を再現するための条件から

$$m_{\psi}^2 \geq (0.6 \text{ TeV})^2$$

ψ はHiggs portal scalar dark matter?

ϕ は未知のスカラー場